Do Now · retrieval

Lesson 070 · Heuristics and Performance Modelling

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is correlation?

Q2

What does a heuristic provide?

Q3

What is an optimisation objective?

Q4

A sorted 32-element candidate set is halved to one candidate. How many halvings are needed?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Define and predict — A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

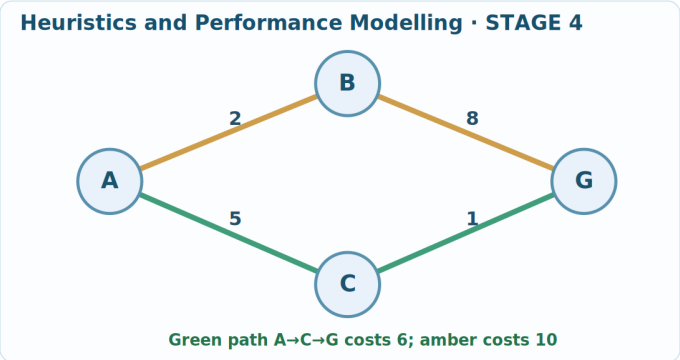
Core idea 2

Explain — A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

Core idea 3

Evaluate — Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

heuristic — Estimate guiding solution search.

greedy — Choose apparently best immediate option.

performance model — Simplified resource/time estimate.

admissible — Heuristic never overestimates remaining optimal cost.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

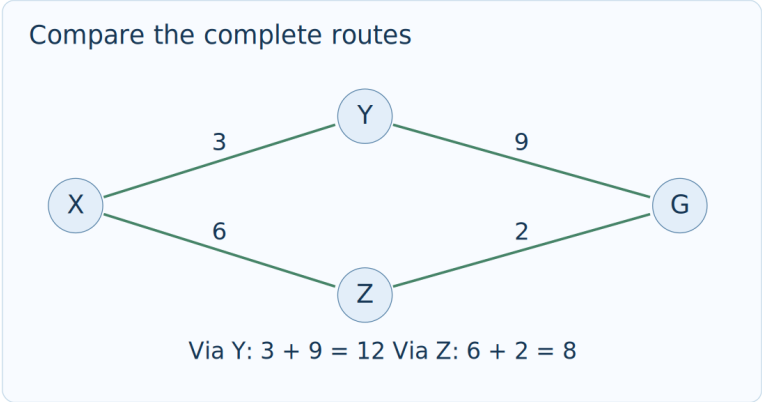
Problem

Choose cheaper complete path: X→Y→G is 3+9; X→Z→G is 6+2.

Follow the worked method

1. Follow the reasoning — Calculate the entire feasible route rather than the next immediate edge.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Follow the reasoning
A performance model is a simplified prediction, not a guaranteed elapsed time.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Compare route A-B-G edges 2+8 against A-C-G 5+1. Calculate totals and select cheaper. [6]

Question 2

Model 1200 jobs processed at 80/s, under constant throughput: estimate duration and one assumption. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate using straight-line distance as heuristic for car routes with one-way roads and congestion. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A local greedy choice always produces an optimal complete path.

Use key vocabulary from memory

1. Explain heuristic in your own words.

2. Explain greedy in your own words.

3. Explain performance model in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.